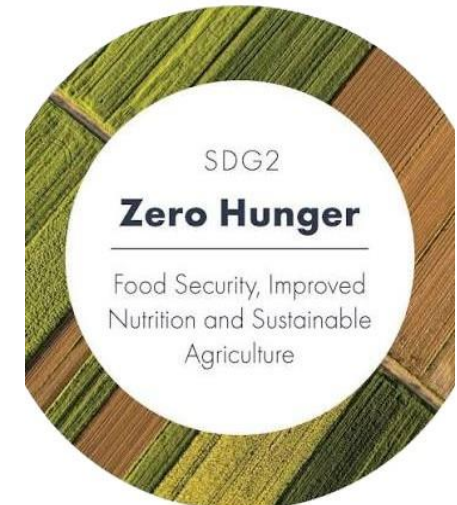


ZERO HUNGER: DIGITAL FOOD-AID SUPPLY CHAIN PLATFORM

UN SUSTAINABLE DEVELOPMENT GOAL 2
(ZERO HUNGER)

Bridging the Gap Between Surplus Food and
Community Need

- **Presenter Name:** Sandip Maity
- **College Name:** Adamas University



PROBLEM STATEMENT & MOTIVATION

Food Insecurity Is Primarily A Coordination Failure Where Surplus Food Goes To Waste While Vulnerable Communities Lack A Unified Digital Tracking System To Bridge The Gap.



KEY CHALLENGES:

- ❑ Massive surplus food goes to waste daily in restaurants, events, and corporate canteens.
- ❑ Vulnerable communities and shelters struggle with food insecurity due to lack of real-time visibility.
- ❑ Fragmented communication (phone calls, ad-hoc messaging) leads to logistical breakdowns and delayed pickups.
- ❑ **OBJECTIVE:** Build a unified, transparent, and scalable digital platform to streamline food rescue operations.

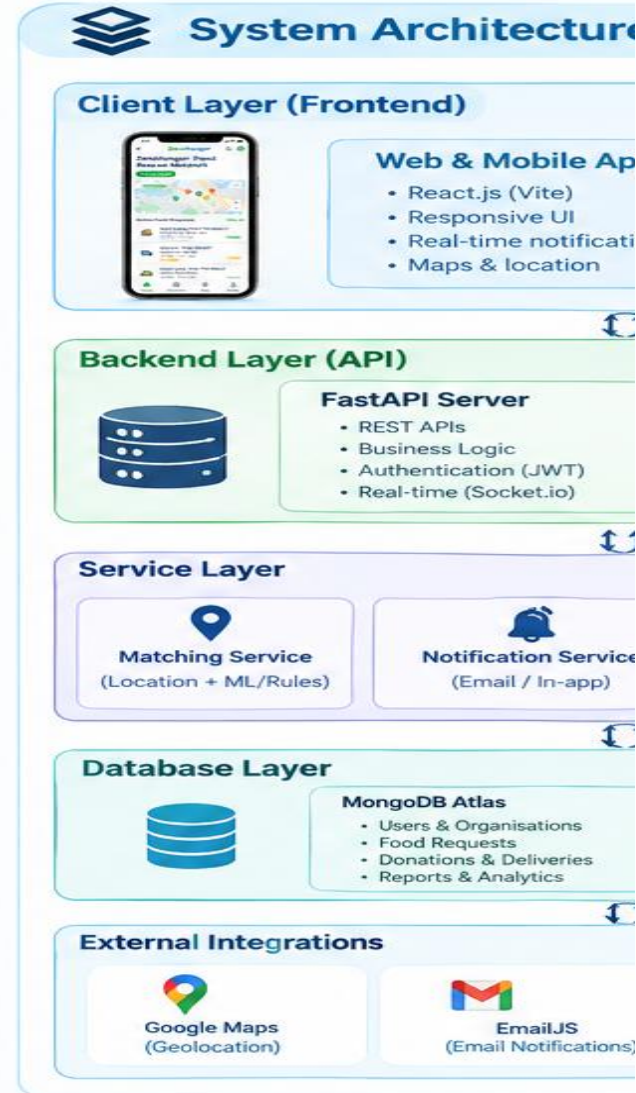
SOLUTION OVERVIEW & FEATURES

A Full-stack Web Application That Digitizes The Entire Food-aid Supply Chain, Connecting Surplus Food Sources With Communities That Need It Most.

Core Features:

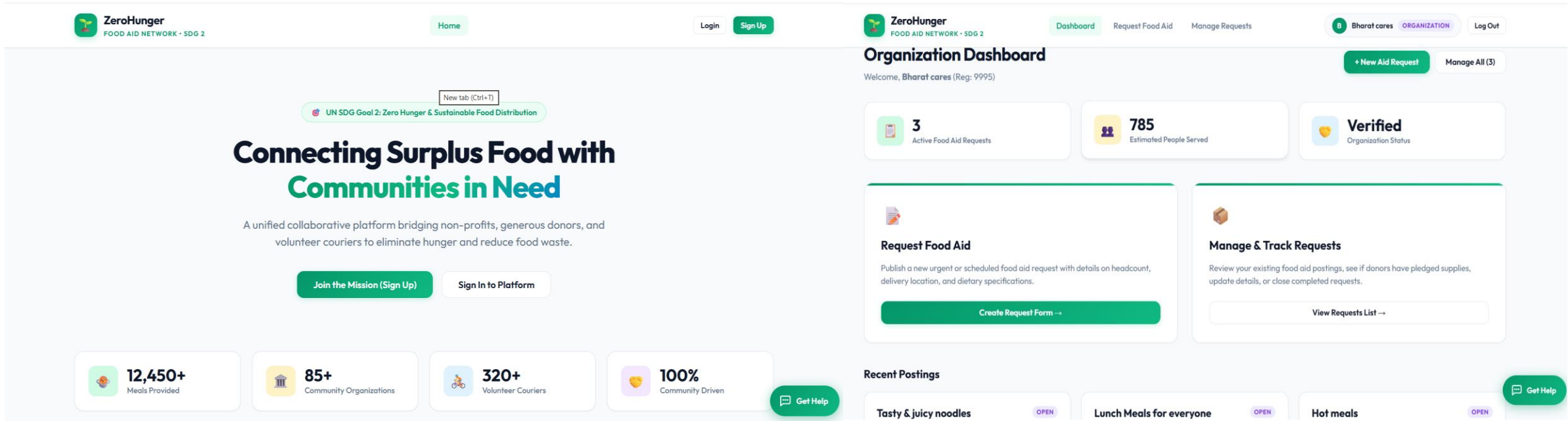
- **Role-Based Access Control (RBAC):** Customized dashboards for Organizers, Donors, Volunteers, and Admins.
- **Real-Time Request Matching:** Instant posting and browsing of food-aid requirements with headcount and location data.
- **End-to-End Tracking:** Live tracking of donation status from pledge to final delivery.
- **Audit & Management:** Admin tools for user verification, account management, and system integrity.





WORKFLOWS & SYSTEM ARCHITECTURE

- Organisations (Shelters, Pantries, NGOs): Post food-aid requests with headcount, location, and date needs.
- Donors (Restaurants, Businesses, Individuals): Browse requests, pledge surplus food, and choose delivery methods.
- Volunteers (Couriers, Drivers): Accept delivery jobs, manage pickups, and coordinate drop-offs.
- Admins (Platform Managers): Oversee registrations, audit system activity, and ensure compliance.
- Technology Stack: Built using React.js, Tailwind CSS, Node.js, Express.js, and MongoDB.



FEASIBILITY & VIABILITY ANALYSIS

Solution(video): [Link](#)

- **Technical Feasibility:** Built using industry-standard web tech (MERN stack), ensuring high modularity, easy local debugging, and smooth cloud deployment capabilities (Vercel + Render).
- **Economic Viability:** Low-cost infrastructure utilizing modern cloud tiers, open-source libraries, and serverless/containerized hosting.
- **Operational Feasibility:** Intuitive UI/UX design reduces the learning curve for non-technical volunteers and community shelter managers.

Request Food Aid

Manage Requests

Lunch Meals for everyone

OPEN REQUEST

Organization:Edunet Foundation

Headcount:200 people

Required By:9/20/2026

Destination: Barasat, North 24 Parganas, West Bengal

Contact Phone:9002838296

Notes:individual

Delete Request

PLATFORM

ROLES

ZeroHunger

FOOD AID NETWORK • SDG 2

Dashboard

Request Food Aid

Manage Requests

B

Bharat cares

ORGANIZATION

Log Out

Organization Dashboard

Welcome, Bharat cares (Reg: 9995)

+ New Aid Request

Manage All (3)

3

Active Food Aid Requests

785

Estimated People Served

Verified

Organization Status

Request Food Aid

Publish a new urgent or scheduled food aid request with details on headcount, delivery location, and dietary specifications.

Create Request Form →

Manage & Track Requests

Review your existing food aid postings, see if donors have pledged supplies, update details, or close completed requests.

View Requests List →

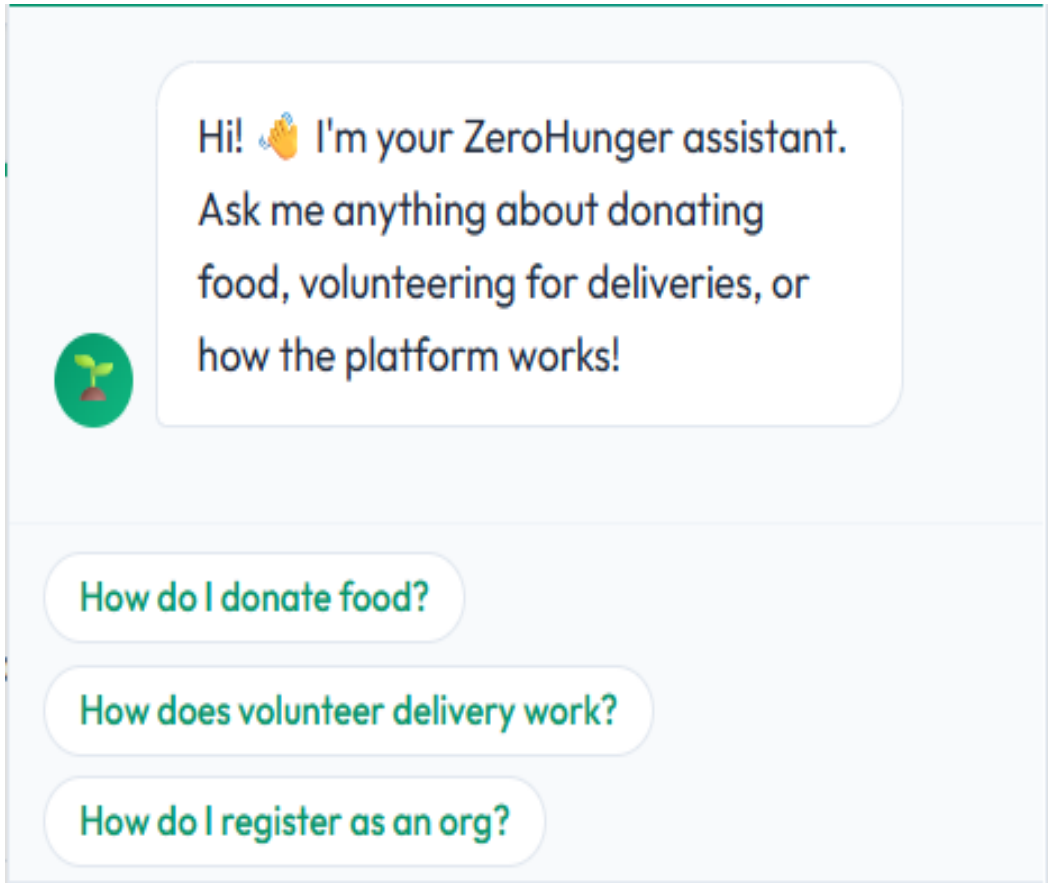
Recent Postings

Get Help

IMPACT STATEMENT & SOCIETAL BENEFITS

Quantifiable Social & Environmental Shifts:

- ❑ **For Hungry Communities:** Reliable, consistent, and safe access to daily nutrition.
- ❑ **For Food Businesses:** Reduced waste disposal costs and streamlined Corporate Social Responsibility (CSR) compliance.
- ❑ **For Volunteers:** A structured, trackable, and gamified way to contribute to society.
- ❑ **For Society:** Lower organic waste in landfills, directly cutting down greenhouse gas (methane) emissions.



FUTURE SCOPE & SCALABILITY

Advanced AI Integration:

- ❑ Predictive demand forecasting to anticipate food surpluses in specific neighborhoods using historical data.
- ❑ Smart route optimization algorithms for volunteers to minimize transit time and fuel consumption.

Platform Expansion:

- ❑ Mobile application wrapper (React Native) for push notifications and GPS tracking.
- ❑ Integration with local municipal food waste management systems.

ZERO HUNGER CHALLENGE

Transforming our food systems to transform our world



All food systems are sustainable from production to consumption



An end to rural poverty double small-scale producer incomes & productivity



Adapt all food systems to eliminate loss or waste of food



Access adequate food & healthy diets, for all people all year round



An end to malnutrition in all its forms

CONCLUSION & RESEARCH REFERENCE

PROJECT(Link)
<https://zero-ruby-eight.vercel.app/login>

GitHub(Link):

<https://github.com/Sandip-Maiti-2023/Zero>

• Summary:

ZeroHunger bridges the gap between digital innovation and social good, turning logistical chaos into an organized ecosystem.

Key Takeaway for Students:

- Think critically about real-world friction.
- Apply scalable software design with purpose.
- Build technology that creates measurable human impact.

References & Research Work:

- UN Sustainable Development Goals (SDG 2 Framework) [Link](#)
- MERN Stack Architectural Patterns & Supply Chain Digitization Research Papers [Link](#)